Parental wealth or parental education: what drives maths results (PISA 2025)

The data only partly support the claim that wealth matters more than education: education is the stronger predictor in 59 of 87 countries, wealth in 28, mostly low-resource ones.

1. Key findings

The hypothesis that parental wealth affects children's results more than parental education was only partly confirmed.

- **On average, parental education has a somewhat stronger link to results than wealth.** With both factors in the model, one standard deviation of parental education corresponds to about 18 maths points, and one standard deviation of home wealth (the HOMEPOS index) to about 12.5. Education is the stronger predictor in 59 of the 87 countries with complete data.
- **In poorer countries the hypothesis holds.** Wealth matters more than education in 28 countries, and 20 of them sit in the third of countries with the fewest home resources: Zambia, Malaysia, Colombia, Ecuador, Brazil, Vietnam, Argentina, Peru, Morocco and others. In the middle and upper thirds, wealth is stronger in only 8 of 58 countries.
- **The two factors reinforce each other rather than compete.** Parental education pays off most in well-off families, while wealth changes results most in families with university-educated parents: among them, the poorest and richest 20% differ by about 51 points.
- **A large part of "wealth" is really cultural capital.** The number of books at home is linked to results more strongly than either parental education or material possessions (TVs, cars, computers and so on).
- **Together the two factors explain only about 10% of score differences within a country.** The other 90% depends on schools, teachers, motivation, ability and chance. That is good news for policy: results are not predetermined by family background.

2. Data and method

The file holds maths results (10 PISA plausible values per student), student weights, parents' education (HISCED, and PAREDINT in years of schooling: 6, 9, 12, 14, 16), the home possessions index HOMEPOS, the composite ESCS index, the number of books at home and five questions about household items. It covers 747,000 students in 89 countries and economies.

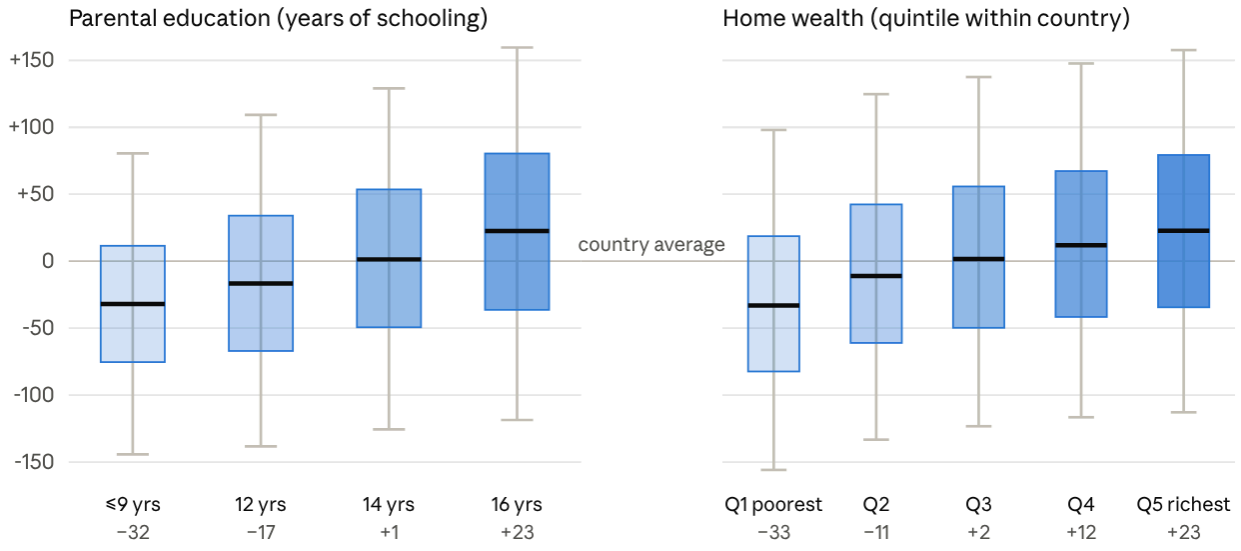
- All calculations were run **separately for each country**, with student weights and separately for each of the 10 plausible values, and the results were then averaged. This is the correct way to work with PISA data.
- Countries count equally, so large countries do not drown out small ones.
- The analysis uses students with known parental education and wealth: 636,000 of 747,000 (85%), from 87 of 89 countries.
- The strength of a factor is measured by standardized regression coefficients and by the share of score differences it explains (R^2).
- Countries are split into thirds by their average HOMEPOS (low, middle, high) as a rough proxy for development, because the file has no GDP data.
- In the cross-tabulation, a student's score is taken relative to the average of their own country.

3. Overall picture

In raw terms the two gaps are almost identical: students from the poorest 20% of homes trail the richest 20% by about 54 points, and children of parents with up to 9 years of schooling trail those with 16 years by about 54 points, roughly two to three school years of learning.

Median scores rise about 55 points from the bottom to the top group, on both measures

Score vs. own country average, points. Box = middle 50% of students; whiskers = 5th to 95th percentile.



Black bar in each box and the number under it = median.

PISA 2025 file · 636,000 students in 87 countries, countries weighted equally, student weights applied

The boxes overlap heavily: many children in the bottom groups outscore many in the top groups, so family background shifts the distribution without deciding an individual child's result.

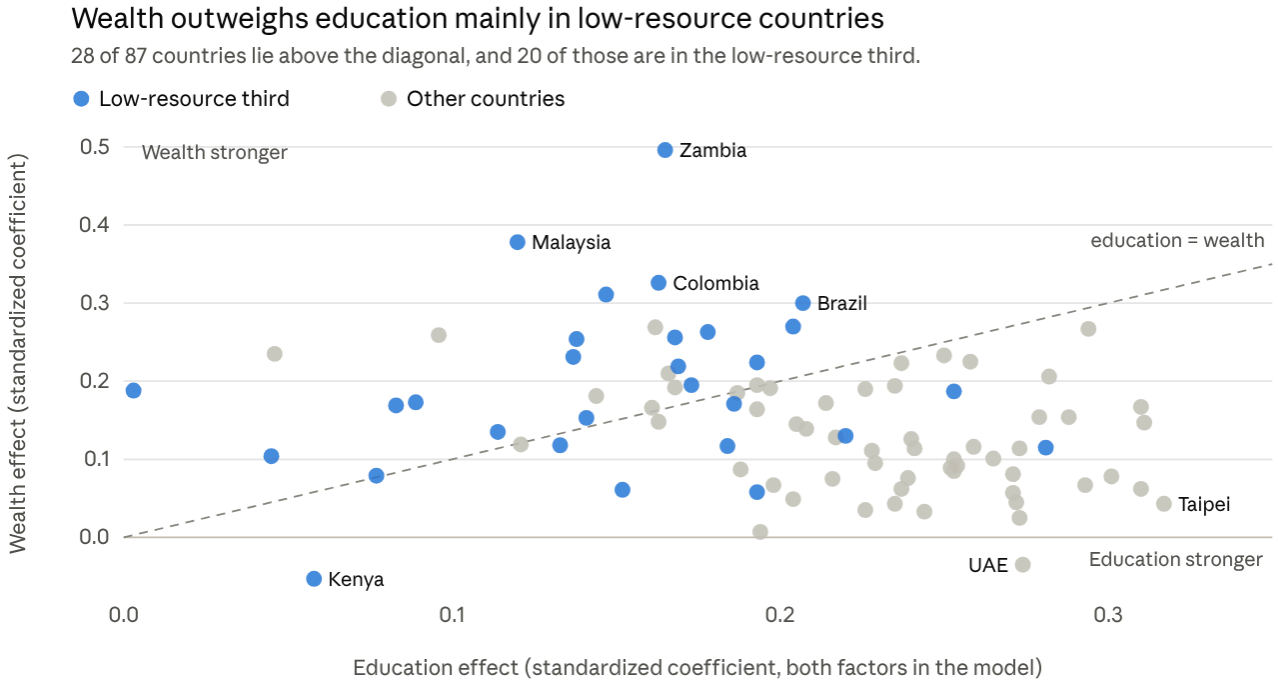
The difference appears when both factors are put in one model and compared within each country:

Average across countries	Parental education	Home wealth (HOMEPOS)
Standardized coefficient, both in the model	0.20	0.15
Maths points per 1 standard deviation	about 18	about 12.5
Share of score differences explained, alone	7.2%	5.7%
Countries where the factor is stronger	59 of 87	28 of 87

Both factors together explain 9.7% of score differences. The share of students below about 420 points (roughly the basic proficiency threshold) is 60% in the poorest fifth and 37% in the richest fifth, against 74% for children of parents with up to 9 years of schooling and 34% for those with 16 years.

4. Country differences

Education has the stronger link with maths results in 59 of 87 countries and wealth in 28, so the answer depends on where you look.



PISA 2025 file · 87 countries with complete data · low-resource third = lowest average home wealth index

In the low-resource third of countries wealth is the stronger factor in 20 of 29 cases (average coefficients 0.19 for wealth against 0.15 for education). In the middle and upper thirds education leads clearly (0.23 against 0.13 and 0.23 against 0.12), and wealth is stronger in only 8 of 58 countries, usually by a small margin (Italy, Austria, New Zealand, the USA). The largest wealth advantages are in Zambia, Malaysia, Colombia, Ecuador and Brazil. Education dominates most strongly in countries such as the UAE, Qatar, Macao, Canada, Taipei and Malta.

Practical reading: **where material conditions at home are uncertain, money and possessions become the main barrier. Where basic conditions are secured for almost everyone, what decides the result is whether the family can help with learning.**

5. What "wealth" really measures

HOMEPOS mixes material possessions with books. Splitting it into its parts (82 countries with complete data) gives:

Factor	Average standardized coefficient
Books at home	0.195
Parental education	0.162
Material possessions (five household-item questions)	0.110

Books are the strongest single factor in 47 countries, parental education in 20, material possessions in 15. So the "effect of wealth" is to a large extent the effect of a **home learning environment**, which correlates with money but is not the same thing.

Correlations between the indicators

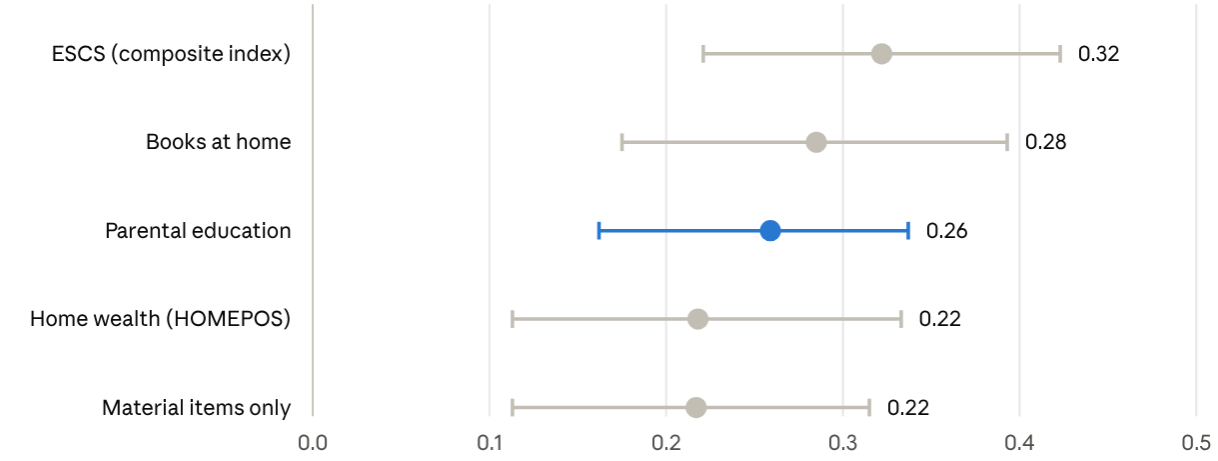
The indicators overlap only partly: parental education and home wealth correlate at 0.34, so they carry largely different information.

Correlation (average across countries)	Parental education	Home wealth (HOMEPOS)	ESCS	Books at home	Material items
Parental education	1.00	0.34	0.79	0.29	0.28
Home wealth (HOMEPOS)	0.34	1.00	0.72	0.43	0.78
ESCS	0.79	0.72	1.00	0.43	0.57
Books at home	0.29	0.43	0.43	1.00	0.34
Material items	0.28	0.78	0.57	0.34	1.00

ESCS is built from parental education and HOMEPOS, which is why it correlates with both, and HOMEPOS is built largely from material items. Next to maths scores:

Books and parental education track maths scores more closely than home wealth

Correlation with maths score. Dot = average across 87 countries; whisker = 10th to 90th percentile of countries.



PISA 2025 file · weighted correlations computed inside each of 87 countries, then averaged · maths = average over 10 plausible values

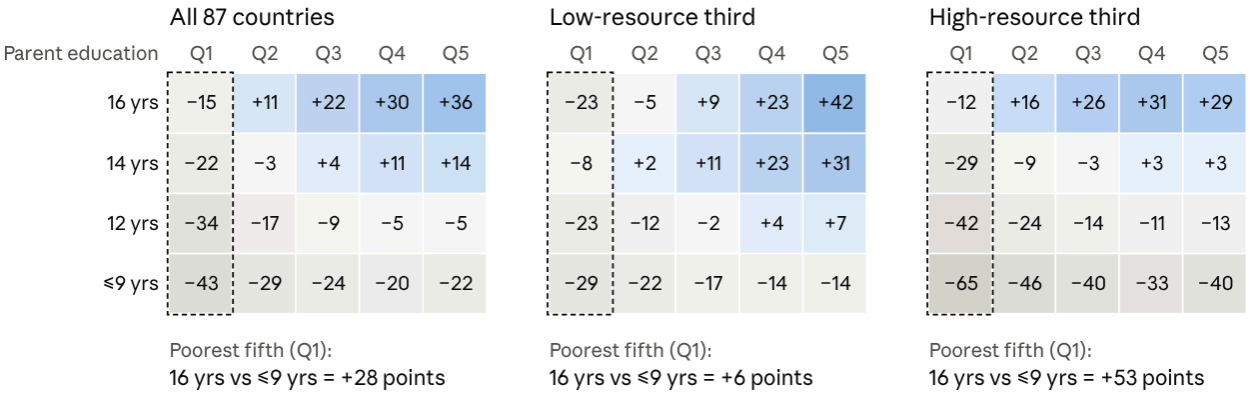
Correlations with maths are moderate, between 0.2 and 0.3 on average, and the range across countries is wide: for parental education the 10th to 90th percentile of countries runs from 0.16 to 0.34, and for home wealth from 0.11 to 0.33. Differences of a few hundredths between indicators are smaller than that spread, so the ranking of education against wealth should not be read as exact in any single country.

6. Education and wealth together

Education and wealth reinforce each other: the best results belong to children of educated and well-off parents, the worst to children of poorly educated and poor ones, a gap of 79 points across all countries.

Education lifts the poorest children far less in low-resource countries: +6 points vs +53

Score vs country average, points. Rows = parental education, columns = home wealth fifth; dashed = poorest fifth.



Q1 = poorest fifth, Q5 = richest, within each country. Blue = above country average, grey = below; darker = larger gap.

PISA 2025 file · average score gap vs own country, equal country weights · low- and high-resource thirds = lowest and highest average home wealth index

What the picture shows:

1. **Higher education partly compensates for poverty.** Across all countries, children of university-educated parents in the poorest fifth trail less than children of parents with up to 9 years of schooling at middle wealth.
2. **Money without education helps little.** Among parents with up to 9 years of schooling, the gap between poor and rich is only 21 points and levels off from the third fifth. Among university-educated parents it is 51 points.
3. **Atypical groups are small.** Poorly educated but richest families make up 1.2% of students, educated but poorest families 5.1%. The largest single group is educated and well-off families, at 13.1%.
4. **In low-resource countries high parental education barely rescues a poor family.** The poorest-fifth children of graduates trail those of parents with up to 9 years of schooling by only about 6 points, against about 53 in high-resource countries. Cells with few students, such as 14 years of schooling in poor homes, are noisy, so read the pattern rather than single cells.

7. Possible explanations

These are hypotheses, not proven causes.

- **Parental education passes on knowledge, language, expectations and the skill of helping with schoolwork.** This works even on modest incomes.
- **Wealth buys time and access:** tutors, clubs, a quiet place to study, internet, books, a better neighbourhood and a stronger school. It matters most where basic conditions cannot be taken for granted.

- **The two are linked** (a correlation of about 0.34 on average across countries): educated parents usually earn more, so part of the "effect of wealth" is the effect of education, and the other way round.
- **Education is tied to family capital more than it seems.** A highly educated parent in a poor family opens the same doors, but they lead to less well-resourced schools and areas.
- **The effect accumulates over time.** Money buys access to quality early education and good schools, which raise the quality of schooling and later income. Educated parents are best placed to use these opportunities.

8. Limitations

- **These are correlations, not causes.** The data do not show that raising parents' income or education would raise children's scores. Hidden factors such as ability and motivation may be passed on within families.
- **Wealth is measured weakly.** HOMEPOS counts household items reported by the 15-year-old student; it is not income or assets, and it includes books. Parental education is coarse, with only five levels (6, 9, 12, 14 and 16 years), which should weaken its apparent effect. Even so, when both were compared on equal terms (9 education categories against 9 wealth groups), education still explained more: 11.2% against 7.4% of score differences, and in 72 of 87 countries.
- **Missing data are not random.** Parental education is missing for 14.6% of students, and these students score on average 40 points below their country's average, while those who answered score 7 points above it. The most disadvantaged children are underrepresented, so real gaps are probably larger than shown.
- **Ukraine is not in the file.** Conclusions about Ukraine need separate data, such as national studies or PISA data if the country took part. Results for countries with similar resources (Moldova, Georgia, Armenia, Romania, Poland, Bulgaria) can serve as a guide, not a replacement.
- **A country's development level is approximated** by its average HOMEPOS, not by GDP.
- The file is named PISA 2025; the analysis used the indicators and values it contains and did not check them against official documentation.

9. Ideas for solutions

The data point to three mechanisms that can be addressed separately: **material conditions** (especially in low-resource countries), the **home learning environment** (books, help with study), which school can compensate for, and **parents' own education**

and involvement. The ideas below follow these directions. All of them are hypotheses to pilot and evaluate, not guaranteed recipes.

A. Remove material barriers

1. **A free basic "learning pack":** textbooks, notebooks, a device and internet access for those who lack them. It matters most in countries where wealth outweighs education.
2. **Free school meals.** They remove hunger as an obstacle to learning and add household income for poor families.
3. **Free clubs and extended-day programmes** (homework support on school premises). They replace what wealthy families buy as clubs and tutors.
4. **Needs-based school funding:** schools with a larger share of poor students get more money per student, not an equal amount for all.

B. Replace the home learning capital that is missing

5. **Small-group tutoring at school** for students who fall behind: short regular sessions of 2–4 students with a trained tutor. This has some of the strongest evidence among school interventions, but it is expensive, so target it at those who need it most.
6. **Books at home and school libraries.** Books at home are the strongest single factor in the data. Book-gift programmes, subscriptions and family reading are cheap and easy to pilot. Books are a marker, not a magic fix, so pair them with reading activities.
7. **Early education (preschool).** Inequality forms before school, so free or subsidized access for children from poor families should have the longest-lasting effect.
8. **Early signals and fast help:** diagnostic tests at the end of primary school and automatic extra support for those who fall behind, not only on request from parents (educated parents ask more often).

C. Work with parents

9. **Programmes for parents with low education:** simple advice on supporting learning (how to check homework, how to talk about school), short formats and local coordinators.
10. **Adult education:** access to evening school, vocational courses and online learning. This is a long-term investment, because parental education is one of the strongest factors.
11. **Transparent school communication:** plain language and consultations for families unfamiliar with the system (choice of clubs, lyceum, admission).

D. System-level decisions

12. **Less school segregation:** transparent admission without hidden selection, later tracking by ability, balanced school catchment areas.
13. **The best teachers in the hardest schools:** pay supplements, mentoring, smaller classes where poverty is concentrated.
14. **Career guidance and "steps up":** scholarships, mentoring and preparation programmes for capable children from poor families and from families with low education.

E. Measurement

15. **A disadvantage indicator as an official metric:** publish school results adjusted for students' social background, so that schools working in hard conditions are not punished and what works becomes visible.
16. **Pilots with impact evaluation** (such as random assignment or comparison of similar schools) before scaling up. PISA data show the problem well but do not say which treatment works.

What to choose when

Situation	Most justified steps
Low-resource country where wealth matters more than education	A1–A4 (pack, meals, clubs, needs-based funding), B5 (tutoring), B7 (preschool)
Middle- or high-resource country where parental education matters more	B5, B6, C9–C11, D12–D13, B8
Limited budget	B6, B8, C11 (cheaper), and selective B5 for those who need it most
Long-term strategy	B7, C10, D12, D14–E16

10. Next steps

- If the problem concerns a specific country, for example Ukraine, it can be compared with its neighbours in this file, or analysed in depth if you provide national data.
- The robustness of the conclusions can be checked with other indicators such as ESCS, or with country-level incomes.
- Any of the ideas can be developed in detail: budget, target group, success indicators and a pilot plan.